

The code is available at <https://github.com/JohnDiamataris/ShannonProduct>.

The code is made up of three files:

1. Generation_Shannon_cMorg.V4.1.py
2. QtPhenmodelmaker.py
3. gen_to_phen.py

and must be in the same directory.

Also a PDF help file is in the github directory

The part of the code that generates the mean, variation and std errors is in the def make_offspring () on line 507.

The part of the code that calculates the joint probabilities are the def get_cross_prob() and def fill_Shannon_prob() on lines 411 and 439

In order to recreate the experiment

Run file Generation_Shannon_cMorg.V4.1.py and on the screen that appears fill in at parent 1 and parent 2 the genotypes V,v;R,r (this is the gene notation under consideration), then press Make Genotypes

On the list generated fill in 1 parent1 at genotype Aa,Bb and 1 parent 2 at genotype aa,bb, leave the rest zero. These are the parent genotypes used in the original Morgan Hunt test cross experiment.

- Fill in the sample size: 500
- Number of samples 1500
- Fill in the crossover probability: 0.125
- Press compute crossover prob. It takes some time for the calculations.
- Do not press Fit Shannon Parameters
- Press autofill Shannon prob. It calculates the joint Shannon probability
- Then press Shannon Product – it calculates and compares the mean of the crossover probability and the Shannon product.